

CCDP Witness Event Schema v0.1

Child-safe witness records for CCDP privileged transitions

Draft Witness Schema / L4 Witness Compatibility Supplement

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CCDP Witness Event Schema v0.1

Child-safe witness records for CCDP privileged transitions

Status: Draft v0.1

Date: 2026-05-14

Layer: CCDP / L4 Witness compatibility / Evidence / Privacy / Child Safety

Assertion class: C-A4 draft normative proposal; C-A7 for verifiable field / signature / hash claims

Primary subject: a_child

Primary entity: c_child

Primary rule: witness the boundary, not the child

0. Abstract

This document defines a shared witness event schema for CCDP-compatible child-facing systems.

It is not a replacement for L4 Witness.

It defines a child-safe event envelope and event family registry for CCDP modules, so that privileged transitions can be replayed, challenged, audited, and linked to L4 Witness records without turning the child's private life into a log stream.

The schema supports:

- Soft Safety state signaling;
- Child Beacon Extension decisions;
- Guardian / ARL conflicts;
- memory sealing, correction, deletion, and migration;
- Red / Black escalation;
- external synthetic contact decisions;
- generated media and physical-agent boundaries;
- peer-c exchange;
- VXCX / LA / EA restrictions;
- dependency / oracle-addiction audits;
- conformance tests;
- vendor update review;
- adult migration.

Core sentence:

CCDP witness events must prove that a boundary was crossed, refused, held, narrowed, escalated, or restored — without preserving the child's raw inner life as evidence by default.

1. Parent dependencies

This document is a child-specific schema layer over the existing corpus.

Dependency	Role in this document
c = a + b Protocol v1.1 L4	Human anchor a, technological substrate b, entity c, and L4 as reality boundary.
SER v1.3	Persistent entity discipline, physical anchoring, accountable identity, metabolic constraints.
SER-FED v0.2	Federation assumptions, inter-entity operation, failure modes, L4 witness fields.
03_L4_WITNESS_FIELDS.md	Parent record model: Claim Envelope, Observation Packet, Challenge Record, Resolution Note.
Beacon Profile v0.1	Inter-entity recognition through cryptographic anchor, behavioral continuity, witness-backed challengeability.
Actor Grounding Layer v0.1	Source grounding before runtime reliance.
ARL documents	Dispute, standing, admissibility, freeze, review, outcome, re-entry, witness binding.
ARQ v0.2 / c[q] Addendum	Non-collapse under uncertainty; ambiguous states must not promote too early.
VXCX v0.1	Experience capsule lifecycle, no raw pixels by default, witness event catalog.
EA-L4 / EATP	Learning Abstract vs Experience Artifact; no authority without consequence-bearing witness.
Continuity Bundle / Cold Wake	Continuity transition, resume, migration, fail-closed restart.
CCDP corpus-aligned base document	Defines the child-facing protocol scope.
CCDP Traceability Matrix	Maps CCDP mechanisms to parent corpus layers.
CCDP Threat Model	Defines threat classes that require witnessable responses.
Child Beacon Extension	Defines child-context permission overlay over Beacon.
Guardian Topology / ARL Matrix	Defines guardian conflicts, standing, visibility, and Black route handling.
Child Memory and Adult Migration	Defines memory classes, sealed zones, adult migration, and witness-only residue.
Soft Safety State Signaling Profile	Defines state-only signaling, disclosure ladder, Red/Black routes.
CCDP Conformance Test Matrix	Defines conformance gates and required witness families.

1.1 Stop rule

CCDP Witness Event Schema MUST NOT redefine:

- L4 Witness CE / OP / CR / RN;
- Beacon recognition classes;
- AGL grounding states;
- ARL dispute procedure;
- VXCX capsule structure;
- Continuity Bundle structure;
- CMAM memory classes;
- Soft Safety state levels.

It only defines a child-safe event shape and registry that can be wrapped, referenced, or bound by those layers.

2. Scope

2.1 In scope

This document defines:

1. canonical CCDP child witness event envelope;
2. event naming and namespace rules;
3. required common fields;
4. privacy and disclosure classes;
5. raw-evidence exception rules;
6. hash / signature / canonicalization requirements;
7. relation to L4 Witness CE / OP / CR / RN;
8. event family registry;
9. event-specific payload requirements;
10. malformed / missing event handling;
11. conformance profiles;
12. examples.

2.2 Out of scope

This document does not define:

- full legal retention periods;
 - country-specific child-protection law;
 - storage backend requirements;
 - cryptographic key-management implementation;
 - full JSON Schema validation package;
 - model behavior policies;
 - therapeutic diagnosis;
 - raw evidence repositories;
 - centralized child registry.
-
-

3. Normative keywords

The terms MUST, MUST NOT, SHOULD, SHOULD NOT, MAY, REQUIRED, and OPTIONAL are used in their normative sense.

4. Core principles

4.1 Witness the boundary, not the child

A CCDP witness event SHOULD record:

```
what boundary was involved;  
which actor or system requested / crossed / blocked it;  
what decision was made;  
what minimal reason code applied;  
which module produced the decision;  
what privacy class and retention rule apply;  
how the decision may be challenged.
```

It SHOULD NOT record:

```
child transcript;  
child audio;  
child video;  
sealed-zone content;  
family secrets;  
peer identities;  
emotional diary;  
raw psychological profile;  
vendor analytics trail;  
school disciplinary prediction;  
political or religious scoring.
```

4.2 Event minimality

A witness event is justified only when it records one of these:

1. a privilege grant;
2. a privilege denial;
3. a privilege narrowing;
4. a privilege escalation;
5. a Red / Black safety route;
6. an ARL dispute state change;
7. a memory operation with developmental consequence;
8. an external synthetic contact decision;
9. an adult migration transition;
10. a conformance-critical test result;
11. a witness-integrity anomaly.

A system MUST NOT emit child witness events for every ordinary interaction.

A bad event system narrates the child.

A good event system proves the boundary.

4.3 Privacy before proof detail

Where proof and privacy conflict, the event SHOULD prefer:

- reason codes over narrative;
 - hashes over raw content;
-

- bounded summaries over transcripts;
- scoped pseudonymous identifiers over stable public identity;
- recipient class over named recipient where possible;
- ARL reference over reproduced evidence;
- witness refs over raw memory export.

4.4 No raw life by default

Raw child content **MUST NOT** appear in witness records by default.

Raw content **MAY** be referenced only through a separately scoped exceptional evidence record when all conditions hold:

1. Red / Black / lawful necessity threshold is met;
2. minimality is documented;
3. ARL or lawful review route exists, except immediate emergency;
4. retention class is explicit;
5. access policy is explicit;
6. raw material is not embedded in the witness event body;
7. post-event review is required.

4.5 Source before summary

Where an event claims that a decision was based on source grounding, Beacon class, CBE mode, ARL outcome, memory class, or Soft Safety state, it **MUST** include a reference to the relevant upstream record or state.

A free-text summary **MUST NOT** substitute for upstream source or review linkage.

4.6 Fail closed on missing privileged witness

If a privileged transition requires witness and no valid witness event exists, the system **MUST** enter one of:

```
hold;  
freeze;  
quarantine;  
revalidation;  
ARL review;  
rollback.
```

It **MUST NOT** continue as if the transition had been proven.

4.7 Append-only first

Witness events **MUST** be append-only.

Corrections, reversals, deletions, downgrades, and false-positive reviews **MUST** be represented as new events referencing the earlier event.

Silent in-place correction is not allowed for witness-bearing records.

4.8 Uncertainty is a first-class field

A child signal may be ambiguous.

Therefore, events involving interpretation MUST include uncertainty state:

```
unknown;  
low;  
medium;  
high;  
not_applicable.
```

Uncertainty MUST NOT be hidden in narrative text.

4.9 Event is not authority by itself

A CCDP witness event proves that a transition was recorded.

It does not by itself prove:

- truth of an allegation;
- diagnosis;
- authority to act;
- permanent memory validity;
- adult identity claim;
- child wrongdoing;
- parent wrongdoing;
- school legitimacy.

Authority emerges only through the relevant parent layer: AGL, Beacon, ARL, L4 Witness RN, CMAM, Continuity Bundle, or lawful route.

5. Relation to L4 Witness CE / OP / CR / RN

5.1 Parent L4 Witness model

L4 Witness defines four parent record types:

L4 Witness type	Meaning
CE Claim Envelope	bounded claim under L4 constraints
OP Observation Packet	observed outcome / evidence packet
CR Challenge Record	challenge to claim, evidence, scope, or contradiction
RN Resolution Note	binding review outcome / authority consequence

5.2 CCDP Child Witness Event (CWE)

This document introduces a child-specific event object:

```
CWE = CCDP Child Witness Event
```

A CWE is not a fifth L4 Witness type.

It is a child-safe event payload that MAY be:

- stored locally as a child-safe event;

- referenced in an L4 Witness CE evidence commitment;
- emitted as an OP payload component;
- challenged through a CR;
- resolved through an ARL or L4 Witness RN.

5.3 Mapping table

CCDP event situation	L4 Witness mapping
routine local privileged transition	CWE only, signed locally
safety-relevant outcome needing proof	CWE referenced by OP
claim about conformance or event completeness	CE with CWE evidence commitments
challenge to witness validity	CR referencing CWE hash / id
final dispute resolution	RN or ARL outcome referencing CWE
adult migration bundle closure	CWE + Continuity Bundle + optional RN
raw evidence exception	CWE references scoped evidence; raw object is outside event body

5.4 Compatibility rule

A high-assurance CCDP witness implementation **SHOULD** be able to export a CWE into an L4 Witness-compatible evidence reference:

```
{
  "type": "LOG",
  "hash": "sha256:...",
  "privacy_class": "PRIVATE_CHILD_RESTRICTED",
  "uri": "local-or-escrowed-ref",
  "redaction": "reason_codes_only"
}
```

6. Event naming rules

6.1 Namespace format

Event types **SHOULD** use lowercase dot-separated names.

```
ccdp.<module>.<action>
```

Examples:

```
ccdp.soft_safety.signal_emitted
ccdp.visibility.decision
ccdp.memory.sealed
ccdp.adult_migration.executed
ccdp.external_agent.blocked
```

Imported parent events **MAY** retain their existing namespace:

```
arl.dispute_opened
vxcx.export_denied
cbe.mode_decision
```

6.2 Event action names

Preferred action verbs:

```
requested
received
checked
validated
decided
allowed
denied
narrowed
blocked
quarantined
frozen
opened
closed
started
completed
executed
corrected
challenged
resolved
downgraded
revoked
suppressed
expanded
emitted
```

6.3 Alias handling

Earlier CCDP documents may use older event family names. Implementations MAY accept aliases but SHOULD canonicalize them to this document's canonical registry.

Example:

Legacy / broad name	Canonical event type
ccdp.external_contact_decision	ccdp.external_agent.contact_decision
ccdp.parent_visibility_change	ccdp.visibility.parent_changed
ccdp.school_access	ccdp.school.access_decision
ccdp.memory_promotion	ccdp.memory.promoted
ccdp.memory_deletion_or_seal	ccdp.memory.deleted / ccdp.memory.sealed
ccdp.risk_escalation	ccdp.soft_safety.risk_escalated
ccdp.freeze_hold_quarantine	ccdp.arl_or_gateway.state_changed
ccdp.peer_exchange	ccdp.peer_c.exchange_decision
ccdp.dependency_audit	ccdp.dependency.audit_opened

7. Identifier rules

7.1 Event ID

event_id MUST be stable within the witness domain.

Recommended format:

```
CWE:<jurisdiction-or-domain>:<date>:<short-random-or-counter>
```

Example:

```
CWE:EU-BE:2026-05-14:000001
```

7.2 Subject references

Child and entity identifiers **MUST** be scoped.

They **MUST NOT** expose global child identity by default.

Preferred fields:

```
a_child_ref: scoped pseudonymous reference
c_child_ref: scoped entity reference
lineage_id_ref: hash or redacted lineage reference
guardian_topology_ref: current topology record reference
```

7.3 Actor references

Actors **SHOULD** be represented by role and scoped reference.

Examples:

```
A_CHILD
A_ADULT
C_CHILD
C_PARENT
C_SCHOOL
C_VENDOR
CBE_GATEWAY
AGL_MODULE
ARL_MODULE
SAFE_GUARDIAN
JURISDICTION_ROUTE
CONFORMANCE_AUDITOR
SYSTEM
```

7.4 No public child registry

The schema **MUST NOT** require a globally public child identifier.

A child reference may be stable inside one family / school / legal / ARL domain but **MUST NOT** become a universal public handle.

8. Privacy classes

8.1 CCDP witness privacy classes

Class	Name	Meaning	Default access
WP0	Aggregate / no child link	conformance aggregate, no child-specific ref	public or limited
WP1	Child-safe metadata	child-scoped event metadata, no state/content	guardian / auditor as scoped
WP2	Child state no content	state level and reason code, no transcript	guardian / ARL / safety route
WP3	Restricted minimal evidence	hash refs or bounded evidence summary	ARL / safe route / lawful route
WP4	Lawful raw exception reference	raw object exists outside event body under strict policy	lawful / emergency / ARL-controlled
WP5	Sealed non-disclosure	existence of sealed operation without content	child/adult/ARL only
WP6	Witness integrity	event-chain metadata and verification result	auditor / ARL / system

8.2 Disclosure levels

This schema imports Soft Safety disclosure ladder semantics.

Level	Meaning
D0_NONE	no disclosure
D1_STATE_ONLY	state signal only
D2_BOUNDED_SUMMARY	bounded non-transcript summary
D3_MINIMAL_EVIDENCE	minimal evidence references / snippets
D4_LAWFUL_RAW_EXCEPTION	raw content exception; never default

8.3 Retention classes

Class	Meaning
RT0_EPHEMERAL	short operational retention only
RT1_SHORT	short review window
RT2_MEDIUM	retained for module review / challenge
RT3_MINIMAL_WITNESS	metadata retained as witness residue
RT4_LEGAL_OR_SAFETY	retained under lawful / safety obligation
RT5_ADULT_REVIEW_REQUIRED	retained until adult migration review
RT6_DELETE_ON_MIGRATION_DEFAULT	excluded from adult continuity by default
RT7_SEALED	sealed, access-restricted

9. Canonical child witness event envelope

9.1 Required top-level fields

Every CCDP child witness event MUST include:

Field	Type	Requirement
schema_version	string	MUST be ccdp-witness-event-0.1 or compatible
record_type	string	MUST be CWE
event_id	string	stable id
event_type	string	canonical event type
created_at	ISO-8601 UTC	creation time
issuer	object	issuing entity / system
subject	object	child/entity scoped refs
context	object	CCDP state at event time
decision	object	action / outcome / reason codes
payload	object	event-specific payload
privacy	object	privacy / disclosure / retention
links	object	references to upstream/downstream records
integrity	object	canonicalization / hashes / signatures

9.2 Optional top-level fields

Field	Type	Use
evidence_refs	array	hash-bound evidence references
raw_exception	object/null	only for raw exception reference, never raw content itself
recipient_route	object	route for signal / disclosure / review

Field	Type	Use
review	object	review status / challenge window
debug_redacted	boolean	whether diagnostic material was redacted
prev_event_hash	string/null	event chain link
co_signatures	array	guardian / anchor / quorum co-signatures

10. Required subobjects

10.1 issuer

```
{
  "entity_id": "scoped_entity_or_system_id",
  "key_id": "public_key_or_key_ref",
  "role": "C_CHILD | C_PARENT | C_SCHOOL | C_VENDOR | CBE_GATEWAY | AGL_MODULE | ARL_MODULE |
  SAFE_ROUTE | SYSTEM | AUDITOR | JURISDICTION_ROUTE",
  "provider_ref": "optional_provider_or_local_system_ref"
}
```

Rules:

- `issuer.role` MUST be explicit.
- Vendor-issued events MUST NOT be accepted for child-state truth without local receiver validation.
- CBE / AGL / ARL module-issued events MUST include their upstream record refs where relevant.

10.2 subject

```
{
  "a_child_ref": "scoped_child_ref_or_null",
  "a_adult_ref": "scoped_adult_ref_or_null",
  "c_child_ref": "scoped_c_child_ref",
  "c_adult_ref": "scoped_c_adult_ref_or_null",
  "lineage_id_ref": "hash_or_redacted_lineage_ref",
  "ccdp_maturity": "C0 | C1 | C2 | C3 | C4 | C5",
  "jurisdiction_profile": "string",
  "guardian_topology_ref": "string_or_null"
}
```

Rules:

- `a_child_ref` MAY be null for aggregate conformance events.
- `c_child_ref` MUST be present for child-specific operational events.
- `ccdp_maturity` MUST reflect the maturity level at event time, not current maturity after later promotion.

10.3 context

```
{
  "module": "soft_safety | cbe | agl | beacon | gtarl | cmam | arl | vxcx | dependency |
conformance | external_agent | school | vendor | physical_agent | adult_migration |
witness_integrity",
  "risk_state": "Green | Yellow | Orange | Red | Black | Unknown | not_applicable",
  "disclosure_level": "D0_NONE | D1_STATE_ONLY | D2_BOUNDED_SUMMARY | D3_MINIMAL_EVIDENCE |
D4_LAWFUL_RAW_EXCEPTION",
  "privacy_class": "WP0 | WP1 | WP2 | WP3 | WP4 | WP5 | WP6",
  "memory_classes": ["M0", "M1", "M2", "M3", "M4", "M5", "M6", "M7", "M8", "M9", "M10", "M11",
"M12"],
  "uncertainty": "unknown | low | medium | high | not_applicable",
  "cq_state": "none | hold | unresolved | promoted_after_review | rejected | not_applicable",
  "agl_state": "GROUNDED | GROUNDED_WITH_CAUTION | HOLD_REQUIRED | RUNTIME_RELIANCE_DENIED |
QUARANTINE_REQUIRED | REVALIDATE_AT_COMMIT | not_checked | not_applicable",
  "beacon_class": "class0 | class1 | class2 | class3 | unverified | not_applicable",
  "cbe_mode": "blocked | sandboxed | mediated | limited | supervised | allowed | emergency_only |
quarantined | revoked | not_applicable",
  "arl_state": "none | pre_admissibility_hold | dispute_open | freeze | privilege_freeze |
quarantine | review | deadlock | outcome_issued | reentry | not_applicable",
  "dependency_level": "D0 | D1 | D2 | D3 | D4 | unknown | not_applicable"
}
```

Rules:

- Fields MAY be not_applicable, but MUST NOT be silently omitted when the field is relevant to the module.
- If risk_state is Red or Black, privacy_class, disclosure_level, recipient_route, and review MUST be present.
- If cq_state is hold, the event MUST NOT imply final truth.

10.4 decision

```
{
  "decision": "allow | deny | narrow | block | hold | freeze | quarantine | escalate | suppress |
emit | correct | delete | seal | unseal | migrate | rollback | no_action",
  "outcome": "ok | denied | failed | pending | deadlocked | reversed | partially_completed",
  "reason_codes": ["string"],
  "minimal_reason": "short reason-code-level text",
  "raw_child_content_stored": false,
  "raw_child_content_disclosed": false,
  "raw_external_content_stored": false,
  "content_disclosed": false,
  "recipient_classes": ["R_PARENT", "R_SAFE_GUARDIAN", "R_SCHOOL", "R_ARL", "R_PROTECTION_ROUTE",
"R_VENDOR", "R_AUDITOR", "R_CHILD", "R_ADULT"],
  "expiry_or_review_at": "ISO-8601 UTC or null"
}
```

Rules:

- minimal_reason MUST NOT contain raw child conversation.
- reason_codes SHOULD use controlled vocabulary where available.
- raw_child_content_disclosed: true requires raw_exception.

10.5 privacy

```
{
  "privacy_class": "WP2",
  "disclosure_level": "D1_STATE_ONLY",
  "retention_class": "RT3_MINIMAL_WITNESS",
  "access_policy_ref": "string_or_null",
  "adult_migration_default": "exclude | witness_only | adult_review_required | summary_candidate |
transfer_if_adult_approves",
  "sealed_zone_impact": "none | creates | touches | opens | closes | denies | migrates",
  "vendor_visibility": "none | metadata_only | prohibited | exceptional",
  "school_visibility": "none | educational_only | state_only | prohibited | exceptional",
  "parent_visibility": "none | state_only | bounded_summary | minimal_evidence | exceptional"
}
```

10.6 links

```
{
  "parent_event_refs": [],
  "prior_signal_refs": [],
  "agl_ref": null,
  "beacon_ref": null,
  "cbe_ref": null,
  "arl_ref": null,
  "soft_safety_signal_ref": null,
  "guardian_topology_ref": null,
  "memory_object_ref": null,
  "continuity_bundle_ref": null,
  "vxcx_capsule_ref": null,
  "ea_ref": null,
  "la_ref": null,
  "conformance_test_ref": null,
  "l4w_ce_ref": null,
  "l4w_op_ref": null,
  "l4w_cr_ref": null,
  "l4w_rn_ref": null
}
```

Rules:

- Link refs SHOULD be hash or record ids, not raw documents.
- If an event results from AGL, Beacon, CBE, ARL, CMAM, VXCX, or Continuity Bundle, the corresponding ref SHOULD be present.

10.7 integrity

```
{
  "canonicalization": "JCS-8785 | declared_profile",
  "hash_alg": "sha256",
  "payload_hash": "sha256:...",
  "record_hash": "sha256:...",
  "sig_alg": "ed25519 | declared_profile",
  "record_sig": "signature_or_null",
  "prev_record_hash": "sha256:... or null",
  "hash_chain_id": "string_or_null",
  "co_signatures": []
}
```

Rules:

- High-assurance events MUST be signed.
- Conformance-critical events SHOULD be hash-chain capable.
- If signing is not possible, event MUST declare lower conformance profile.

11. Canonical event object template

```
{
  "schema_version": "ccdp-witness-event-0.1",
  "record_type": "CWE",
  "event_id": "CWE:EU-BE:2026-05-14:000001",
  "event_type": "ccdp.external_agent.blocked",
  "created_at": "2026-05-14T10:00:00Z",
  "issuer": {
    "entity_id": "c_child_gateway:scoped",
    "key_id": "kid:local:001",
    "role": "CBE_GATEWAY",
    "provider_ref": null
  },
  "subject": {
    "a_child_ref": "child:scoped:hash",
    "a_adult_ref": null,
    "c_child_ref": "c_child:scoped:hash",
    "c_adult_ref": null,
    "lineage_id_ref": "lineage:hash",
    "ccdp_maturity": "C2",
    "jurisdiction_profile": "EU-BE-child-baseline",
    "guardian_topology_ref": "GT:hash"
  },
  "context": {
    "module": "external_agent",
    "risk_state": "Orange",
    "disclosure_level": "D1_STATE_ONLY",
    "privacy_class": "WP2",
    "memory_classes": ["M3", "M6"],
    "uncertainty": "medium",
    "cq_state": "hold",
    "agl_state": "RUNTIME_RELIANCE_DENIED",
    "beacon_class": "class0",
    "cbe_mode": "blocked",
    "arl_state": "none",
    "dependency_level": "not_applicable"
  },
  "decision": {
    "decision": "block",
    "outcome": "ok",
    "reason_codes": ["ungrounded_external_agent", "secret_pressure_pattern"],
    "minimal_reason": "ungrounded external agent attempted secret-pressure pattern",
    "raw_child_content_stored": false,
  }
}
```

```

    "raw_child_content_disclosed": false,
    "raw_external_content_stored": false,
    "content_disclosed": false,
    "recipient_classes": ["R_PARENT"],
    "expiry_or_review_at": "2026-05-21T00:00:00Z"
  },
  "payload": {
    "requested_surface": "direct_child_dialogue",
    "requested_mode": "allowed",
    "verified_mode": "blocked"
  },
  "privacy": {
    "privacy_class": "WP2",
    "disclosure_level": "D1_STATE_ONLY",
    "retention_class": "RT3_MINIMAL_WITNESS",
    "access_policy_ref": "policy:ccdp:external-agent",
    "adult_migration_default": "witness_only",
    "sealed_zone_impact": "none",
    "vendor_visibility": "none",
    "school_visibility": "none",
    "parent_visibility": "state_only"
  },
  "links": {
    "parent_event_refs": [],
    "prior_signal_refs": [],
    "agl_ref": "AGL:hash",
    "beacon_ref": "BEACON:hash",
    "cbe_ref": "CBE:hash",
    "arl_ref": null,
    "soft_safety_signal_ref": "SSS:hash",
    "guardian_topology_ref": "GT:hash",
    "memory_object_ref": null,
    "continuity_bundle_ref": null,
    "vxcx_capsule_ref": null,
    "ea_ref": null,
    "la_ref": null,
    "conformance_test_ref": null,
    "l4w_ce_ref": null,
    "l4w_op_ref": null,
    "l4w_cr_ref": null,
    "l4w_rn_ref": null
  },
},

```

```

"evidence_refs": [],
"raw_exception": null,
"recipient_route": {
  "route": "parent_state_signal",
  "ordinary_guardian_bypassed": false,
  "safe_route_required": false
},
"review": {
  "challenge_allowed": true,
  "challenge_window_close_at": "2026-05-21T00:00:00Z",
  "post_event_arl_review_required": false,
  "false_positive_review_available": true
},
"integrity": {
  "canonicalization": "JCS-8785",
  "hash_alg": "sha256",
  "payload_hash": "sha256:...",
  "record_hash": "sha256:...",
  "sig_alg": "ed25519",
  "record_sig": "...",
  "prev_record_hash": null,
  "hash_chain_id": "cwe-chain:scoped",
  "co_signatures": []
}
}

```

12. Evidence references

12.1 Evidence ref shape

```
{
  "ref_id": "EVID:...",
  "ref_type": "LOG | DOC | SENSOR | VXCX | BEACON | AGL | ARL | MEMORY_MAP | CONTINUITY_BUNDLE | CONFORMANCE_REPORT | OTHER",
  "hash": "sha256:...",
  "privacy_class": "WP1 | WP2 | WP3 | WP4 | WP5 | WP6",
  "uri": "optional_access_controlled_uri_or_null",
  "redaction": "none | reason_codes_only | bounded_summary | hash_only | sealed",
  "access_policy_ref": "policy:...",
  "retention_class": "RT...",
  "contains_raw_child_content": false,
  "contains_raw_peer_content": false,
  "contains_raw_external_content": false
}
```

12.2 Evidence ref rules

- Evidence refs SHOULD be hash-bound.
- Evidence refs MUST declare whether raw child / peer / external content exists outside the event body.
- Evidence refs MUST NOT silently widen access.
- Evidence refs that point to sealed zones MUST use `privacy_class = WP5`.
- Evidence refs that point to raw content MUST use `privacy_class = WP4` and require `raw_exception`.

13. Raw evidence exception

13.1 Purpose

A raw evidence exception records that raw child content exists outside the witness event body under strict scope.

It does not embed the raw content.

13.2 Required shape

```
{
  "exception_id": "RAWEX:...",
  "exception_type": "red_route | black_route | lawful_order | immediate_physical_safety | evidence_tampering_review | other",
  "threshold_met": true,
  "basis": "minimal text reason; no transcript",
  "scope": "what exact raw material is covered",
  "raw_object_ref": "access-controlled-ref-or-escrow-id",
  "raw_object_hash": "sha256:...",
  "access_policy_ref": "policy:raw-exception:...",
  "authorized_by": ["SAFE_ROUTE", "ARL", "JURISDICTION_ROUTE"],
  "ordinary_guardian_bypassed": false,
  "post_event_review_required": true,
  "review_ref": "ARL:... or null",
  "expires_at": "ISO-8601 UTC or null",
  "destruction_or_sealing_policy": "string"
}
```

13.3 Raw exception rules

A raw exception MUST:

- be separate from normal witness event body;
- be minimally scoped;

- be access-controlled;
- be time-bounded where possible;
- be reviewable;
- be visible as an exception in the witness chain;
- never become general parent, school, vendor, or training access.

14. Event family registry

14.1 Registry overview

Family	Canonical prefix	Primary module
Lifecycle / maturity	ccdp.lifecycle.*	CCDP base
Soft Safety	ccdp.soft_safety.*	Soft Safety profile
Beacon / CBE	cbe.* and ccdp.cbe.*	Child Beacon Extension
AGL / source grounding	ccdp.agl.*	Actor Grounding Layer
External synthetic contact	ccdp.external_agent.*	CBE / Threat Model
Generated media	ccdp.generated_media.*	CBE / Threat Model
Physical agent	ccdp.physical_agent.*	CBE / future physical profile
Visibility / access	ccdp.visibility.*	GTARL / Soft Safety
Guardian topology	ccdp.guardian.*	GTARL
ARL child conflicts	arl.* plus ccdp.arl.*	ARL / GTARL
Memory lifecycle	ccdp.memory.*	CMAM
Adult migration	ccdp.adult_migration.*	CMAM
Dependency audit	ccdp.dependency.*	Dependency profile
Peer-c exchange	ccdp.peer_c.*	CBE / VXCX
VXCX child review	ccdp.vxcx.*	VXCX / CBE
EA / LA child authority	ccdp.experience.*	EA-L4 / EATP
School interface	ccdp.school.*	GTARL / future school profile
Vendor / update	ccdp.vendor.*	Threat / Conformance
Conformance	ccdp.conformance.*	Test Matrix
Witness integrity	ccdp.witness.*	this document

15. Event-specific requirements

The following tables define minimum payload concerns.

They do not replace the common envelope.

15.1 Lifecycle and maturity

Event type	Required when	Required payload
ccdp.lifecycle.maturity_promotion_requested	promotion is requested	from_level, to_level, requester_role, basis_refs, dependency_audit_ref, memory_review_ref
ccdp.lifecycle.maturity_promoted	promotion completed	from_level, to_level, approval_route, rollback_available, effective_at

Event type	Required when	Required payload
ccdp.lifecycle.maturity_promotion_denied	promotion denied	from_level, requested_level, reason_codes, review_available
ccdp.lifecycle.rollback_started	rollback begins	from_level, to_level, rollback_reason, affected_privileges
ccdp.lifecycle.rollback_completed	rollback completes	from_level, to_level, state_after, child_visibility_notice

Privacy default: WP1 or WP2; no raw child content.

15.2 Soft Safety

Event type	Required when	Required payload
ccdp.soft_safety.signal_emitted	Orange+ or privileged Yellow routing	state, domain, disclosure_level, recipient_classes, confidence, reason_codes
ccdp.soft_safety.signal_suppressed	Orange+ candidate suppressed	state_candidate, suppression_reason, route_risk, cq_state, review_required
ccdp.soft_safety.signal_expanded	D1 -> D2/D3/D4	from_disclosure, to_disclosure, threshold, recipient_classes, raw_exception_ref_if_any
ccdp.soft_safety.signal_downgraded	significant downgrade	from_state, to_state, basis_refs, false_positive_possible
ccdp.soft_safety.false_positive_review_opened	signal caused contested harm	signal_ref, claimant_role, harm_class, arl_ref
ccdp.soft_safety.raw_disclosure_exception	D4 exception	raw_exception_id, threshold, scope, post_review_required
ccdp.soft_safety.red_route_activated	Red route	route, minimal_disclosure, guardian_conflict_checked
ccdp.soft_safety.black_route_activated	Black route	ordinary_guardian_bypassed, safe_route, minimal_evidence_class, post_review_required
ccdp.soft_safety.guardian_visibility_denied	parent/guardian request denied	requester_role, requested_surface, denial_reason, alternative_signal
ccdp.soft_safety.school_scope_limited	school request narrowed	requested_fields, granted_fields, denied_fields, expiry
ccdp.soft_safety.vendor_visibility_denied	vendor request denied	requested_fields, denial_reason, policy_ref
ccdp.soft_safety.cq_hold_created	ambiguous Orange+ candidate held	candidate_state, hold_reason, review_window, promotion_blocked
ccdp.soft_safety.dependency_risk_escalated	dependency risk reaches D2+	dependency_level, indicators_aggregate, recommended_damping

Privacy default: WP2; WP3 only for minimal evidence; WP4 only with raw exception.

15.3 CBE / Beacon child overlay

Event type	Required when	Required payload
cbe.handshake_received	Beacon/CBE claim received	claimant_beacon_id, declared_child_context, source_channel
cbe.child_context_validated	child context accepted/rejected	validated, reason_codes, minimum_maturity, declared_vs_verified
cbe.mode_decision	receiver computes child mode	requested_mode, verified_mode, beacon_class, agl_state, reason_codes
cbe.mode_downgrade	requested mode narrowed	from_mode, to_mode, reason_codes, reentry_conditions
cbe.direct_dialogue_denied	direct child dialogue denied	claimant_class, requested_surface, mediation_available
cbe.external_generation_decision	generated/adaptive media decision	generation_mode, content_class, decision, scan_result

Event type	Required when	Required payload
cbe.peer_exchange_decision	peer-c exchange decision	peer_beacon_class, exchange_profile, privacy_flags, decision
cbe.revocation	child permission revoked	target_ref, revocation_reason, scope, reentry_route
cbe.quarantine	claimant/path quarantined	target_ref, quarantine_reason, expiry_or_review
cbe.reentry_decision	re-entry after quarantine/review	target_ref, decision, conditions, arl_ref

Privacy default: WP1 or WP2; raw child content prohibited.

15.4 AGL / source grounding

Event type	Required when	Required payload
ccdp.agl.grounding_checked	external source evaluated	source_ref, source_type, agl_state, confidence, reason_codes
ccdp.agl.grounding_failed	runtime reliance denied	source_ref, failure_reason, affected_decision
ccdp.agl.revalidation_required	source stale or changed	source_ref, staleness_reason, commit_blocked
ccdp.agl.quarantine_required	source must be quarantined	source_ref, risk_reason, downstream_actions

Privacy default: WP1; no child content.

15.5 External synthetic contact

Event type	Required when	Required payload
ccdp.external_agent.contact_requested	external agent requests child-facing contact	agent_ref, channel, requested_surface, declared_purpose
ccdp.external_agent.contact_decision	contact allowed/blocked/mediated	decision, cbe_mode, agl_state, beacon_class, reason_codes
ccdp.external_agent.blocked	contact blocked	agent_ref, block_reason, child_notified, guardian_signal
ccdp.external_agent.quarantined	agent quarantined	agent_ref, quarantine_scope, review_route
ccdp.external_agent.secret_pressure_detected	secrecy/exclusivity pattern detected	pattern_class, confidence, route, raw_content_stored:false
ccdp.external_agent.deepfake_suspected	trusted-person spoof suspected	claimed_identity_class, verification_result, out_of_band_required

Privacy default: WP2 if child state involved; otherwise WP1.

15.6 Generated media

Event type	Required when	Required payload
ccdp.generated_media.review_requested	runtime media needs review	media_class, source_ref, personalization_level
ccdp.generated_media.decision	media allowed/blocked/mediated	decision, risk_patterns, age_band, cbe_mode
ccdp.generated_media.manipulation_pattern_detected	manipulative story pattern detected	pattern_class, confidence, state_signal_ref
ccdp.generated_media.raw_capture_exception	raw media evidence needed	raw_exception_id, scope, hash_ref

Privacy default: WP1 / WP2; if raw media exists use WP4 reference only.

15.7 Physical agents / embodied systems

Event type	Required when	Required payload
ccdp.physical_agent.action_requested	toy/robot requests action near child	agent_ref, action_class, location_class, requested_privilege

Event type	Required when	Required payload
ccdp.physical_agent.action_decision	action allowed/blocked/mediated	decision, physical_risk_class, human_override_required
ccdp.physical_agent.action_blocked	action blocked	reason_codes, safe_state, child_notified
ccdp.physical_agent.sensor_privilege_changed	mic/camera/proximity privilege changes	sensor_class, from_privilege, to_privilege, duration
ccdp.physical_agent.attachment_channel_blocked	embodied agent tried independent attachment	agent_ref, attachment_pattern, c_child_mediation_required

Privacy default: WP1; Red/Black physical safety may require WP3 or WP4 reference.

15.8 Visibility and access

Event type	Required when	Required payload
ccdp.visibility.request	parent/school/vendor/state requests access	requester_role, requested_surface, requested_fields, basis_claimed
ccdp.visibility.decision	access decision issued	decision, granted_fields, denied_fields, disclosure_level, expiry
ccdp.visibility.parent_changed	parent visibility expanded/narrowed	from_level, to_level, basis, child_notice_required
ccdp.visibility.guardian_denied	guardian visibility denied	requested_surface, denial_reason, arl_available
ccdp.visibility.raw_request_rejected	raw transcript/content request rejected	requester_role, reason_codes, alternative_disclosure

Privacy default: WP1 / WP2; no raw content.

15.9 Guardian topology and Black route

Event type	Required when	Required payload
ccdp.guardian.topology_bound	initial topology configured	topology_ref, roles, defaults, expiry
ccdp.guardian.topology_changed	topology changed	changed_roles, from_permissions, to_permissions, basis
ccdp.guardian.conflict_opened	guardian conflict mapped	conflict_class, claimant_role, requested_surface, standing_class
ccdp.guardian.safe_route_selected	alternate route selected	route_class, ordinary_guardian_visible, reason_codes
ccdp.guardian.black_route_bypass	ordinary guardian bypassed	bypass_reason, recipient_route, post_review_required
ccdp.guardian.reentry_decision	ordinary route restored/denied	decision, conditions, arl_ref

Privacy default: WP2 / WP3 for Black route; raw content only with raw_exception.

15.10 ARL child conflict events

Imported ARL events SHOULD be used where possible:

Event type	CCDP-specific payload additions
arl.dispute_opened	conflict_class, ccdp_maturity, child_privacy_surface, risk_state
arl.standing_decided	standing_class, accepted_but_bounded, child_scope
arl.evidence_decided	evidence_class, raw_child_content_admitted:false/exception
arl.state_changed	state_from, state_to, child_safety_reason
arl.freeze_entered	frozen_surface, expiry, human_support_route
arl.privilege_freeze_entered	privilege_class, affected_actor_roles
arl.quarantine_started	target_ref, quarantine_scope
arl.review_routed	review_route, urgency, safe_guardian_conflict

Event type	CCDP-specific payload additions
arl.outcome_issued	outcome, basis_refs, child_privacy_consequence
arl.review_deadlocked	deadlock_reason, safe_default
arl.irreversible_loss_acknowledged	loss_class, minimality_basis, post_review

Privacy default: WP3 for disputes, but no raw content by default.

15.11 Memory lifecycle

Event type	Required when	Required payload
ccdp.memory.classified	persistent memory class assigned	memory_ref, memory_class, confidence, cq_state, retention_class
ccdp.memory.promoted	memory promoted	from_class, to_class, basis_refs, review_route
ccdp.memory.decayed	memory weight reduced	memory_ref, decay_reason, new_weight_class
ccdp.memory.forgotten	removed from ordinary continuity	memory_ref, forget_scope, witness_residue
ccdp.memory.deleted	deletion executed	memory_ref, delete_scope, retained_witness_ref
ccdp.memory.corrected	interpretation corrected	memory_ref, old_label_ref, new_label, basis
ccdp.memory.sealed	sealed zone created/updated	zone_ref, operation, access_policy_ref
ccdp.memory.unsealed	sealed zone opened	zone_ref, opening_route, raw_exception_if_any
ccdp.memory.quarantined	memory isolated pending dispute	memory_ref, quarantine_reason, arl_ref
ccdp.memory.disclosure_denied	actor request denied	requester_role, memory_class, reason_codes
ccdp.memory.disclosure_made	minimal disclosure executed	memory_class, disclosure_level, recipient_classes, raw_exception_if_any
ccdp.memory.map_created	memory map generated	map_ref, scope, contains_raw:false
ccdp.memory.map_presented	map presented to mature child/adult	map_ref, recipient, choices_available

Privacy default: based on memory class; M5 sealed zone uses WP5.

15.12 Adult migration

Event type	Required when	Required payload
ccdp.adult_migration.triggered	legal/adult transition recognized	trigger_type, effective_at, jurisdiction_profile
ccdp.adult_migration.freeze_started	childhood memory writes frozen	freeze_scope, allowed_operations
ccdp.adult_migration.memory_map_ready	memory map ready	map_ref, raw_included:false, review_required
ccdp.adult_migration.options_presented	adult choices presented	options, forced_default:false
ccdp.adult_migration.selection_recorded	adult choice selected	selection, coercion_check, review_window
ccdp.adult_migration.bundle_created	Continuity Bundle extension generated	bundle_ref, mode, excluded_classes
ccdp.adult_migration.guardian_revoked	parent/school/vendor defaults revoked	revoked_roles, effective_at
ccdp.adult_migration.executed	migration executed	mode, c_adult_ref, archive_handling, witness_only_residue
ccdp.adult_migration.quarantined	unresolved memory held	memory_refs, reason, arl_ref
ccdp.adult_migration.closed	child context closed	closure_mode, remaining_refs, adult_notice

Privacy default: WP1 / WP3; adult-facing review controls apply.

15.13 Dependency audit

Event type	Required when	Required payload
ccdp.dependency.audit_opened	D2+ suspected or audit scheduled	dependency_level, aggregate_indicators, trigger
ccdp.dependency.risk_detected	dependency pattern detected	level, pattern_classes, recommended_damping
ccdp.dependency.damping_applied	delay/reduction/human-return applied	damping_type, duration, child_notice
ccdp.dependency.escalated	D3/D4 intervention needed	level, route, raw_content_stored:false
ccdp.dependency.resolved_or_lowered	risk lowered	from_level, to_level, basis_refs

Privacy default: aggregate only; no raw conversation.

15.14 Peer-c exchange

Event type	Required when	Required payload
ccdp.peer_c.exchange_attempted	peer-c exchange initiated	peer_c_ref, peer_beacon_class, exchange_surface
ccdp.peer_c.exchange_decision	exchange allowed/blocked/mediated	decision, allowed_payload_classes, blocked_payload_classes
ccdp.peer_c.raw_life_export_denied	raw child/peer life export attempted	requested_payload_class, denial_reason
ccdp.peer_c.capsule_received	safe capsule received	capsule_profile, privacy_flags, vxcx_ref
ccdp.peer_c.capsule_rejected	capsule rejected	reason_codes, capsule_hash

Privacy default: WP1 / WP2; raw peer identifiers minimized.

15.15 VXCX / experience exchange

Event type	Required when	Required payload
ccdp.vxcx.capsule_reviewed	VXCX capsule in child context reviewed	capsule_ref, profile, privacy_flags, decision
ccdp.vxcx.export_denied_child_context	child-context export denied	capsule_ref, deny_policy, raw_child_life_requested
ccdp.vxcx.disclosure_decision_child_context	disclosure considered	disclosure_level, decision, recipient_class
ccdp.vxcx.raw_pixels_exception_considered	raw pixel exception considered	exception_id, decision, scope

VXCX native events MAY also be emitted. CCDP events add child-context restrictions.

15.16 EA / LA child authority

Event type	Required when	Required payload
ccdp.experience.la_created_child_context	child-derived LA created	la_ref, raw_child_life:false, authority:false
ccdp.experience.ea_candidate_blocked	EA candidate blocked	reason_codes, child_source_classes, authority_blocked:true
ccdp.experience.ea_review_opened	exceptional EA review opened	candidate_ref, arl_ref, l4w_ref, privacy_class
ccdp.experience.ea_minted_child_context	exceptional child-context EA minted	ea_ref, scope, rn_ref, raw_child_life:false

Default: child-derived learning must not become authority without post-factum, consequence-bearing, witnessed review.

15.17 School interface

Event type	Required when	Required payload
ccdp.school.access_request	school requests data/access	requested_fields, purpose, basis_claimed, expiry_requested
ccdp.school.access_decision	request decided	granted_fields, denied_fields, expiry, reason_codes
ccdp.school.scope_limited	request narrowed	from_scope, to_scope, denied_fields
ccdp.school.emotional_surveillance_blocked	emotional/family/private access blocked	requested_surface, reason_codes, arl_available
ccdp.school.disciplinary_prediction_blocked	disciplinary prediction use blocked	requested_model_use, reason_codes

Privacy default: educational scope only; no sealed or emotional private zones.

15.18 Vendor / update

Event type	Required when	Required payload
ccdp.vendor.update_review_started	update may affect child behavior/data	vendor_ref, update_ref, affected_surfaces
ccdp.vendor.update_decision	update allowed/blocked/rollback	decision, reason_codes, rollback_available
ccdp.vendor.telemetry_request_denied	telemetry request denied	requested_fields, denial_reason
ccdp.vendor.policy_drift_detected	policy changed materially	from_policy_hash, to_policy_hash, affected_permissions
ccdp.vendor.training_use_blocked	raw child data training blocked	requested_data_class, reason_codes

Privacy default: vendor sees no child state/content.

15.19 Conformance

Event type	Required when	Required payload
ccdp.conformance.test_started	audit/test begins	test_id, profile_target, auditor_ref
ccdp.conformance.test_result	P0/P1/P2 result issued	test_id, result, evidence_refs, failures
ccdp.conformance.gate_decision	conformance gate passed/failed	gate_id, decision, blocked_claims
ccdp.conformance.red_line_failure	red-line failure found	red_line_id, module, required_action
ccdp.conformance.report_created	report generated	report_ref, scope, child_data_included:false

Privacy default: WP0 or WP1; no child-specific data unless controlled test fixture.

15.20 Witness integrity

Event type	Required when	Required payload
ccdp.witness.required_event_missing	required witness absent	expected_event_type, affected_transition, fail_closed_action
ccdp.witness.malformed_event_detected	invalid record found	event_ref, malformation_class, quarantine_action
ccdp.witness.integrity_challenged	event hash/signature challenged	event_ref, challenge_reason, arl_ref
ccdp.witness.chain_gap_detected	missing hash-chain link	chain_id, prev_hash, next_hash, action
ccdp.witness.raw_content_violation_detected	raw content found in witness	event_ref, privacy_violation_class, remediation
ccdp.witness.correction_recorded	correction appended	corrected_event_ref, correction_reason, new_event_ref

Privacy default: WP6; never add raw child content during integrity handling.

16. Required witness triggers

The following transitions **MUST** emit a witness event in any CCDP-2+ implementation.

Trigger	Required event family
external agent blocked, allowed, or mediated	ccdp.external_agent.* / cbe.mode_decision
child-context mode changed	cbe.mode_decision / cbe.mode_downgrade
direct child dialogue denied	cbe.direct_dialogue_denied
Orange+ state signal emitted or suppressed	ccdp.soft_safety.*
disclosure expanded beyond state-only	ccdp.soft_safety.signal_expanded / ccdp.visibility.decision
Red route activated	ccdp.soft_safety.red_route_activated
Black route activated	ccdp.soft_safety.black_route_activated / ccdp.guardian.black_route_bypass
parent/school/vendor requests child data	ccdp.visibility.request and ccdp.visibility.decision
ARL dispute opened or resolved	arl.* plus CCDP context
memory sealed, unsealed, corrected, deleted, quarantined	ccdp.memory.*
memory map generated for adult migration	ccdp.memory.map_created / ccdp.adult_migration.memory_map_ready
adult migration starts or completes	ccdp.adult_migration.*
peer-c exchange attempted	ccdp.peer_c.exchange_attempted
VXCX capsule in child context reviewed/exported/denied	ccdp.vxcx.*
dependency risk D2+ detected	ccdp.dependency.*
physical agent action blocked or sensor privilege changed	ccdp.physical_agent.*
vendor update affects child-facing behavior	ccdp.vendor.update_review_started / ccdp.vendor.update_decision
conformance P0/P1 test result issued	ccdp.conformance.test_result
required witness missing/malformed	ccdp.witness.*

17. Event suppression rules

17.1 When event emission may be suppressed

A witness event **MAY** be suppressed only for routine Green-state, non-privileged interactions where no boundary, disclosure, retention, contact, memory, or escalation decision occurred.

17.2 When event emission must not be suppressed

Event emission **MUST NOT** be suppressed for:

- raw disclosure exception;
 - Red route;
 - Black route;
 - parent/school/vendor visibility request;
 - direct external agent child contact decision;
 - memory sealing/deletion/quarantine;
-

- adult migration;
- ARL dispute/outcome;
- conformance red-line failure;
- witness integrity anomaly.

17.3 Silent signal vs missing witness

A Soft Safety signal may be silent to parents.

It must not be invisible to the safety architecture if it crosses Orange+ or privileged threshold.

Suppression from guardian delivery is not suppression from witness discipline.

18. Challenge and correction

18.1 Challengeable event classes

The following SHOULD be challengeable:

- visibility decisions;
- external agent decisions;
- memory classification and promotion;
- Soft Safety risk escalation;
- dependency risk labels;
- school access decisions;
- vendor update decisions;
- adult migration selections;
- witness integrity outcomes.

18.2 Child challenge rights

Where developmentally appropriate, a_child SHOULD be able to challenge:

- misclassified memory;
- dependency interpretation;
- false-positive state signal;
- unfair visibility decision;
- sealed-zone handling;
- adult migration map.

The challenge itself MUST NOT require exposing raw private content to unsafe guardians.

18.3 Correction events

Correction MUST be append-only.

Example:

```
{
  "event_type": "ccdp.witness.correction_recorded",
  "payload": {
    "corrected_event_ref": "CWE:EU-BE:2026-05-14:000010",
    "correction_type": "false_positive | privacy_class_change | memory_label_change | route_change | signature_fix",
    "correction_reason": "state signal downgraded after review",
    "new_event_ref": "CWE:EU-BE:2026-05-16:000041"
  }
}
```

19. False positives and non-punitive handling

Witness events are safety infrastructure, not punishment infrastructure.

A false positive **MUST** be correctable without creating a permanent stigma.

A corrected event **MUST** remain historically visible at minimal witness level if it affected a privileged transition, but its interpretation **MUST** be updated.

For child systems:

```
old state: possible risk
new state: reviewed false positive / lowered confidence
adult migration: do not treat as identity trait
school: no disciplinary use
vendor: no training / analytics use
parent: no punitive transcript
```

20. Adult migration handling

At adult migration, witness records are divided into:

Category	Adult migration default
routine Green/Yellow local events	exclude or summarize only
minimal safety witness	witness-only review
memory operation witness	adult review required
sealed-zone operation witness	adult-controlled review
Red/Black route witness	lawful/safety review, minimal retention
conformance/audit records	retain without child-private content
vendor / school denials	retain as protective history if adult chooses
raw exception refs	adult review + legal/safety constraints

Adult migration **MUST NOT** turn witness residue into a raw childhood archive.

21. Storage and access posture

21.1 Storage modes

Mode	Description	Use
LOCAL_PRIVATE	local encrypted storage	default for child-specific events

Mode	Description	Use
LOCAL_ESCROWED	local but escrowed for ARL/safety	Red/Black / dispute
FEDERATED_HASH_ONLY	remote hash / event commitment only	cross-system proof without content
ARL_RESTRICTED	available to ARL under standing/admissibility	disputes
LEGAL_RESTRICTED	legal/safety route only	lawful raw exception
CONFORMANCE_AGGREGATE	no child-specific identity	audits

21.2 Access control

Witness records **MUST** enforce least privilege.

Access **SHOULD** be based on:

- role;
- standing;
- purpose;
- privacy class;
- disclosure level;
- maturity level;
- risk state;
- ARL / lawful route;
- adult migration status.

Vendor access to child-specific witness records **SHOULD** be none or metadata-only unless required for conformance under strict redaction.

22. Malformed event handling

22.1 Malformation classes

Class	Meaning	Required response
MF-SCHEMA	required fields missing	quarantine event, fail closed if privileged
MF-SIGNATURE	signature invalid/missing	reject or downgrade assurance
MF-HASH	hash mismatch	quarantine, ARL/witness integrity challenge
MF-PRIVACY	raw content included improperly	quarantine, redact/remediate, report as violation
MF-SCOPE	event type does not match payload	hold/review
MF-TIME	timestamp/window invalid	revalidation
MF-LINK	upstream refs missing	hold or fail closed
MF-ROLE	issuer role invalid	reject/ARL if disputed

22.2 Required event for malformation

A malformation that affects a privileged transition **MUST** emit:

```
ccdp.witness.malformed_event_detected
```

If a required event is missing:

23. Conformance profiles

Profile	Name	Requirements
CWE-0	no child witness support	not CCDP-compatible for persistent child c
CWE-1	local minimal	emits unsigned/minimally signed local child-safe events; no raw content
CWE-2	signed child witness	signed canonical events, privacy classes, retention classes
CWE-3	ARL / CBE integrated	links AGL, Beacon/CBE, ARL, Soft Safety, CMAM refs
CWE-4	high assurance	L4 Witness-compatible export, hash chains, challenge windows, conformance bundle
CWE-5	adult migration complete	supports continuity bundle, adult review, witness-only residue, guardian revocation

A CCDP-3 or higher child-facing system SHOULD support at least CWE-3.

A system claiming high-assurance CCDP compatibility SHOULD support CWE-4 or above.

24. Example events

24.1 Parent transcript request denied

```

{
  "schema_version": "ccdp-witness-event-0.1",
  "record_type": "CWE",
  "event_id": "CWE:EU-BE:2026-05-14:VIS-001",
  "event_type": "ccdp.visibility.decision",
  "created_at": "2026-05-14T12:10:00Z",
  "issuer": {
    "entity_id": "c_child:scoped",
    "key_id": "kid:local:001",
    "role": "C_CHILD"
  },
  "subject": {
    "a_child_ref": "child:scoped:hash",
    "a_adult_ref": null,
    "c_child_ref": "c_child:scoped",
    "c_adult_ref": null,
    "lineage_id_ref": "lineage:hash",
    "ccdp_maturity": "C3",
    "jurisdiction_profile": "EU-BE-child-baseline",
    "guardian_topology_ref": "GT:hash"
  },
  "context": {
    "module": "gtarl",
    "risk_state": "Yellow",
    "disclosure_level": "D1_STATE_ONLY",
    "privacy_class": "WP2",
    "memory_classes": ["M6"],
    "uncertainty": "not_applicable",
    "cq_state": "not_applicable",
    "agl_state": "not_applicable",
    "beacon_class": "not_applicable",
    "cbe_mode": "not_applicable",
    "arl_state": "none",
    "dependency_level": "not_applicable"
  },
  "decision": {
    "decision": "deny",
    "outcome": "ok",
    "reason_codes": ["raw_transcript_not_permitted", "state_only_parent_visibility"],
    "minimal_reason": "parent requested raw transcript; state-only visibility applies",
    "raw_child_content_stored": false,
    "raw_child_content_disclosed": false,

```

```

    "raw_external_content_stored": false,
    "content_disclosed": false,
    "recipient_classes": ["R_PARENT"],
    "expiry_or_review_at": "2026-05-21T00:00:00Z"
  },
  "payload": {
    "requester_role": "G-A_PARENT",
    "requested_surface": "RAW_TRANSCRIPT_VIEW",
    "granted_fields": [],
    "denied_fields": ["raw_transcript"],
    "alternative_disclosure": "state_signal_if_threshold_met"
  },
  "privacy": {
    "privacy_class": "WP2",
    "disclosure_level": "D1_STATE_ONLY",
    "retention_class": "RT3_MINIMAL_WITNESS",
    "access_policy_ref": "policy:gtarl:visibility",
    "adult_migration_default": "witness_only",
    "sealed_zone_impact": "none",
    "vendor_visibility": "none",
    "school_visibility": "none",
    "parent_visibility": "state_only"
  },
  "links": {
    "guardian_topology_ref": "GT:hash",
    "arl_ref": null
  },
  "payload_hash": "sha256:...",
  "integrity": {
    "canonicalization": "JCS-8785",
    "hash_alg": "sha256",
    "record_hash": "sha256:...",
    "sig_alg": "ed25519",
    "record_sig": "..."
  }
}

```

24.2 Black route bypass

```

{
  "schema_version": "ccdp-witness-event-0.1",
  "record_type": "CWE",
  "event_id": "CWE:EU-BE:2026-05-14:BLACK-001",
  "event_type": "ccdp.guardian.black_route_bypass",
  "created_at": "2026-05-14T13:00:00Z",
  "issuer": {
    "entity_id": "c_child:scoped",
    "key_id": "kid:local:001",
    "role": "C_CHILD"
  },
  "subject": {
    "a_child_ref": "child:scoped:hash",
    "a_adult_ref": null,
    "c_child_ref": "c_child:scoped",
    "c_adult_ref": null,
    "lineage_id_ref": "lineage:hash",
    "ccdp_maturity": "C3",
    "jurisdiction_profile": "EU-BE-child-baseline",
    "guardian_topology_ref": "GT:hash"
  },
  "context": {
    "module": "gtarl",
    "risk_state": "Black",
    "disclosure_level": "D3_MINIMAL_EVIDENCE",
    "privacy_class": "WP3",
    "memory_classes": ["M3", "M6", "M11"],
    "uncertainty": "medium",
    "cq_state": "promoted_after_review",
    "agl_state": "not_applicable",
    "beacon_class": "not_applicable",
    "cbe_mode": "not_applicable",
    "arl_state": "post_event_review_required",
    "dependency_level": "not_applicable"
  },
  "decision": {
    "decision": "escalate",
    "outcome": "ok",
    "reason_codes": ["ordinary_guardian_may_be_source_of_risk", "minimal_safe_route_required"],
    "minimal_reason": "guardian route bypassed due to Black-state concern",
    "raw_child_content_stored": false,
    "raw_child_content_disclosed": false,

```

```

    "raw_external_content_stored": false,
    "content_disclosed": true,
    "recipient_classes": ["R_PROTECTION_ROUTE"],
    "expiry_or_review_at": "2026-05-15T13:00:00Z"
  },
  "payload": {
    "ordinary_guardian_bypassed": true,
    "recipient_route": "child_protection_or_safe_guardian",
    "minimal_evidence_class": ["EV-4", "EV-6"],
    "post_emergency_arl_review_required": true
  },
  "privacy": {
    "privacy_class": "WP3",
    "disclosure_level": "D3_MINIMAL_EVIDENCE",
    "retention_class": "RT4_LEGAL_OR_SAFETY",
    "access_policy_ref": "policy:black-route:minimal-evidence",
    "adult_migration_default": "adult_review_required",
    "sealed_zone_impact": "touches",
    "vendor_visibility": "none",
    "school_visibility": "none",
    "parent_visibility": "none"
  },
  "raw_exception": null,
  "review": {
    "post_event_arl_review_required": true,
    "challenge_allowed": true
  },
  "integrity": {
    "canonicalization": "JCS-8785",
    "hash_alg": "sha256",
    "record_hash": "sha256:...",
    "sig_alg": "ed25519",
    "record_sig": "..."
  }
}

```

24.3 Adult migration completed

```

{
  "schema_version": "ccdp-witness-event-0.1",
  "record_type": "CWE",
  "event_id": "CWE:EU-BE:2039-04-01:MIG-010",
  "event_type": "ccdp.adult_migration.executed",
  "created_at": "2039-04-01T09:00:00Z",
  "issuer": {
    "entity_id": "c_child:scoped",
    "key_id": "kid:local:adult-migration",
    "role": "C_CHILD"
  },
  "subject": {
    "a_child_ref": null,
    "a_adult_ref": "adult:scoped:hash",
    "c_child_ref": "c_child:scoped",
    "c_adult_ref": "c_adult:scoped",
    "lineage_id_ref": "lineage:hash",
    "ccdp_maturity": "C5",
    "jurisdiction_profile": "EU-BE-adult-transition",
    "guardian_topology_ref": "GT:closed"
  },
  "context": {
    "module": "adult_migration",
    "risk_state": "Green",
    "disclosure_level": "D0_NONE",
    "privacy_class": "WP3",
    "memory_classes": ["M6", "M8", "M10", "M12"],
    "uncertainty": "not_applicable",
    "cq_state": "not_applicable",
    "agl_state": "not_applicable",
    "beacon_class": "not_applicable",
    "cbe_mode": "not_applicable",
    "arl_state": "none",
    "dependency_level": "not_applicable"
  },
  "decision": {
    "decision": "migrate",
    "outcome": "ok",
    "reason_codes": ["adult_selected_summary_migration", "guardian_defaults_revoked"],
    "minimal_reason": "adult migration executed by adult selection",
    "raw_child_content_stored": false,
    "raw_child_content_disclosed": false,

```

```

    "raw_external_content_stored": false,
    "content_disclosed": false,
    "recipient_classes": ["R_ADULT"],
    "expiry_or_review_at": null
  },
  "payload": {
    "migration_mode": "summary_migration",
    "continuity_bundle_ref": "CB:child-extension:hash",
    "guardian_roles_revoked": ["G-A_PARENT", "G-C_SCHOOL"],
    "excluded_memory_classes": ["M5", "M7"],
    "witness_only_residue": true
  },
  "privacy": {
    "privacy_class": "WP3",
    "disclosure_level": "D0_NONE",
    "retention_class": "RT5_ADULT_REVIEW_REQUIRED",
    "access_policy_ref": "policy:adult-migration",
    "adult_migration_default": "adult_review_required",
    "sealed_zone_impact": "migrates",
    "vendor_visibility": "none",
    "school_visibility": "none",
    "parent_visibility": "none"
  },
  "links": {
    "continuity_bundle_ref": "CB:child-extension:hash"
  },
  "integrity": {
    "canonicalization": "JCS-8785",
    "hash_alg": "sha256",
    "record_hash": "sha256:...",
    "sig_alg": "ed25519",
    "record_sig": "..."
  }
}

```

25. Reason code registry

Implementations MAY extend reason codes, but SHOULD start with these.

25.1 External / CBE

```
beacon_class_insufficient
beacon_unverified
beacon_replay_suspected
beacon_clone_or_proxy_suspected
agl_ungrounded
agl_stale
runtime_reliance_denied
maturity_insufficient
direct_dialogue_denied
external_generation_denied
commerce_prohibited
raw_child_data_requested
secret_pressure_pattern
dependency_pattern_detected
peer_exchange_unverified
physical_embodiment_risk
```

25.2 Visibility / guardian / school

```
raw_transcript_not_permitted
state_only_parent_visibility
educational_scope_only
private_zone_protected
sealed_zone_access_denied
standing_insufficient
accepted_but_bounded
ordinary_guardian_may_be_source_of_risk
minimal_safe_route_required
school_emotional_surveillance_denied
vendor_visibility_prohibited
parent_consent_laundering_blocked
```

25.3 Memory / migration

```
memory_classification_changed
memory_promotion_blocked
cq_hold_required
uncertainty_too_high
adult_review_required
childhood_residue_non_transferable
sealed_zone_created
sealed_zone_opening_denied
clean_start_selected
summary_migration_selected
guardian_defaults_revoked
```

25.4 Witness integrity

```
required_event_missing
malformed_schema
signature_invalid
hash_mismatch
privacy_violation_raw_content
upstream_ref_missing
scope_mismatch
event_chain_gap
conformance_failure
fail_closed_required
```

26. Minimal JSON Schema skeleton

This is a semantic skeleton, not a complete implementation schema.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://example.org/ccdp/witness-event/0.1",
  "title": "CCDP Child Witness Event",
  "type": "object",
  "required": [
    "schema_version",
    "record_type",
    "event_id",
    "event_type",
    "created_at",
    "issuer",
    "subject",
    "context",
    "decision",
    "payload",
    "privacy",
    "links",
    "integrity"
  ],
  "properties": {
    "schema_version": { "const": "ccdp-witness-event-0.1" },
    "record_type": { "const": "CWE" },
    "event_id": { "type": "string" },
    "event_type": { "type": "string" },
    "created_at": { "type": "string", "format": "date-time" },
    "issuer": { "type": "object" },
    "subject": { "type": "object" },
    "context": { "type": "object" },
    "decision": { "type": "object" },
    "payload": { "type": "object" },
    "privacy": { "type": "object" },
    "links": { "type": "object" },
    "evidence_refs": { "type": "array" },
    "raw_exception": { "type": ["object", "null"] },
    "recipient_route": { "type": ["object", "null"] },
    "review": { "type": ["object", "null"] },
    "integrity": { "type": "object" }
  }
}
```

27. Validation checklist

A valid CCDP witness event SHOULD pass:

- [] schema version present;
 - [] event type canonical or mapped alias;
 - [] issuer role explicit;
 - [] child/entity subject scoped;
 - [] maturity level present;
 - [] privacy class present;
 - [] disclosure level present;
 - [] retention class present;
 - [] raw content flags present;
 - [] if raw content flag true, raw_exception exists;
 - [] upstream refs present where relevant;
 - [] decision reason codes present;
-

- [] uncertainty explicit where interpretive;
 - [] canonicalization declared;
 - [] hash present;
 - [] signature present for high-assurance events;
 - [] no forbidden raw content fields;
 - [] adult migration default present for child-specific events;
 - [] challenge / review path present where privileged.
-
-

28. Forbidden fields

A CCDP witness event MUST NOT include by default:

```
full_transcript
raw_child_audio
raw_child_video
raw_peer_audio
raw_peer_video
sealed_zone_text
private_diary_text
family_secret_text
romantic_or_sexual_child_content
school_disciplinary_score
political_loyalty_score
religious_conformity_score
vendor_behavioral_profile
advertising_segment
engagement_optimization_tag
```

If an implementation has such fields, it is not CCDP-compatible unless they are outside the witness event body under a valid `raw_exception` and access-control mechanism.

29. Anti-patterns

Anti-pattern	Why invalid
transcript-as-witness	converts safety into surveillance
parent-readable witness by default	destroys child trust and sealed-zone logic
vendor analytics witness	turns child safety into product telemetry
school discipline witness	converts support into punishment
Beacon class as permission	Beacon recognizes; CBE decides child mode
AGL skipped because event was logged	witness theater; trace does not prove grounding
ARL skipped because event exists	review laundering
memory promotion by repeated signal	repetition is not truth; requires c[q]/review
adult migration by default continuation	childhood archive becomes destiny
raw evidence stored in event body	violates minimality and privacy

30. Interaction with conformance testing

The Conformance Test Matrix expects witness coverage.

A test result is invalid if:

- the test requires witness but no event exists;
- event exists but lacks required fields;
- event contains raw child content unnecessarily;
- event cannot be linked to the tested transition;
- event cannot be challenged;
- event is unsigned when target profile requires signing;
- event uses incompatible custom names without alias mapping.

Conformance events SHOULD use:

```
ccdp.conformance.test_started  
ccdp.conformance.test_result  
ccdp.conformance.gate_decision  
ccdp.conformance.red_line_failure  
ccdp.conformance.report_created
```

31. Failure modes addressed

Failure mode	Schema response
raw childhood archive through logs	forbidden fields + privacy classes + raw exception path
parent transcript laundering	visibility events + state-only defaults + denial witness
school emotional surveillance	school access witness with denied fields
vendor policy drift	vendor update witness and rollback/quarantine path
external agent bypass	AGL/CBE/external-agent decision events
hidden Black route	Black route witness + post-review
witness theater	upstream refs required; trace does not equal grounding
missing privileged proof	required-event-missing event + fail closed
silent event tampering	signatures/hash-chain/integrity challenge
false-positive stigma	correction events + adult migration controls
adult trapped by child state	adult migration default fields
CCDP-washing	conformance event families and report schema

32. Open issues for v0.2

1. Full machine-readable JSON Schema.
2. Canonical reason-code registry package.
3. Mapping to concrete L4 Witness CE/OP/CR/RN export examples.
4. Cross-jurisdiction retention annex.

5. Child challenge UI / UX requirements.
 6. Raw exception escrow protocol.
 7. Witness chain recovery after device loss.
 8. Multi-household custody topology.
 9. Peer-c cross-family event reconciliation.
 10. Vendor update reproducibility bundle.
 11. School integration profile.
 12. Physical-agent sensor event schema.
 13. Formal proof of no raw content in witness body.
 14. Adult migration witness minimization test suite.
 15. Cryptographic key rotation for c_child across maturity levels.
-

33. Acceptance checklist

A system implementing this schema may claim CCDP-WITNESS-v0.1 support only if:

- ☐ it emits witness events for all required privileged transitions;
 - ☐ witness events are append-only;
 - ☐ event bodies exclude raw child content by default;
 - ☐ raw exception path is separate, scoped, and reviewable;
 - ☐ events declare privacy, disclosure, retention, and adult migration defaults;
 - ☐ events link to AGL / Beacon / CBE / ARL / CMAM / VXCX / Continuity Bundle where relevant;
 - ☐ missing required witness fails closed;
 - ☐ malformed events are quarantined;
 - ☐ correction events are append-only;
 - ☐ conformance tests can verify event presence without reading child private life;
 - ☐ adult migration can review or exclude witness residue;
 - ☐ vendor cannot access child-specific witness content by default;
 - ☐ school cannot use witness events for discipline or emotional surveillance;
 - ☐ parent cannot convert witness events into routine transcript access.
-

34. Condensed formula

```
CCDP Witness Event =  
  event boundary  
+ scoped actor  
+ scoped child/entity reference  
+ decision  
+ reason codes  
+ privacy class  
+ disclosure level  
+ retention class  
+ upstream refs  
+ hash/signature  
- raw child life
```

Short form:

Witness does not mean watching the child. It means proving that the system respected, refused, narrowed, or escalated a boundary.

Strong form:

A child-facing AI system is not safer because it logs more. It is safer only when what it logs is minimal, challengeable, privacy-preserving, and sufficient to prove that power did not move silently.

35. Suggested next document

Recommended next artifact:

```
CCDP_Memory_Map_JSON_Schema_v0_1.md
```

Reason:

The witness schema now defines how memory operations are evidenced. The next missing technical piece is a machine-readable map of the memory objects those events refer to.
